

1. The end point detection by Volhard titration of Ag^+ with Cl^- ions is commonly characterized by:
 - A) Formation of soluble colored complex FeSCN^{42}
 - B) Formation of a colored precipitate
 - C) Adsorption of a colored indicator on the precipitate
 - D) Turbidity of a solution mixture
2. In the titration of a mixture containing I^- , Br^- , Cl^- ions with AgNO_3 , the order of halides salts precipitation is (K_{sp} : $\text{AgI} = 8.3 \times 10^{-17}$, $\text{AgBr} = 5.0 \times 10^{-13}$, and $\text{AgCl} = 1.8 \times 10^{-10}$):
 - A) AgBr , AgCl then AgI
 - B) AgI , AgBr then AgCl
 - C) AgCl , AgI then AgBr
 - D) AgI , AgCl then AgBr
3. A 25.00 mL of 0.10 M I^- was titrated with 0.05000 M Ag^+ . $[\text{Ag}^+]$ after the addition of 10.0 mL $[\text{Ag}^+]$ is:
 - A) 1.41×10^{-3}
 - B) 1.62×10^{-8}
 - C) 7.58×10^{-3}
 - D) 1.45×10^{-15}
4. $[\text{Ag}^+]$ at the equivalence point is:
 - A) 1.41×10^{-8}
 - B) 9.11×10^{-9}
 - C) 6.92×10^{-7}
 - D) 1.38×10^{-6}
5. $[\text{Ag}^+]$ after the addition of 52.0 mL Ag^+ is:
 - A) 1.29×10^{-3}
 - B) 1.62×10^{-8}
 - C) 6.92×10^{-7}
 - D) 6.60×10^{-3}
6. A weak acid has a pH of 3.05 at 0.25 M concentration. Calculate K_a for this acid:
 - A) 5.94×10^{-3}

- B) 5.29×10^{-6}
- C) 3.17×10^{-6}
- D) 1.33×10^{-4}

7. What is the pH of 0.05 M NaF solution? ($K_a(\text{HF}) = 6.8 \times 10^{-4}$, $K_w = 1.0 \times 10^{-14}$):

- A) 7.93
- B) 9.83
- C) 4.14
- D) 8.08

8. What is the pH of 0.005 M $\text{Ca}(\text{OH})_2$ solution?

- A) 12.30
- B) 12.0
- C) 1.70
- D) 2.00

9. How many milliliters of 0.626 M KOH should be added to 10.0 g of BH (223.29 g/mol) to give a pH of 7.4 ($\text{p}K_a = 7.48$)?

- A) 32.5
- B) 17.0
- C) 15.2
- D) 16.3

10. What is the pH of 100 mL of a 1.0 M acetic acid and 1.0 M sodium acetate buffer ($\text{p}K_a = 4.74$) after the addition of 0.02 mol of NaOH?

- A) 4.74
- B) 9.26
- C) 4.56
- D) 4.92

11. The ionic strength of 0.02 M MgCl_2 is:

- A) 0.05 M
- B) 0.01 M
- C) 0.15 M

- D) 0.30 M

12. Which of the following statements is NOT correct regarding the activity coefficient?

- A) It approaches unity for very dilute solutions (ionic strength = 0)
- B) The activity coefficient of Na^+ increases as the ionic strength of the solution decreases
- C) For a solution of ionic strength = 0.01, the activity coefficient of Ca^{2+} is larger than Na^+
- D) It equals unity for neutral species

13. 0.05 M of a weak base (Bb) has a pH = 9.5. Calculate K_a for its conjugate acid (BH^+) ($K_w = 1.0 \times 10^{-14}$):

- A) 2.0×10^{-8}
- B) 5.0×10^{-10}
- C) 2.0×10^{-18}
- D) 1.8×10^{-5}

14. What is the solubility of PbCl_2 in a saturated solution of H_2O ? ($K_{sp} = 1.7 \times 10^{-5}$):

- A) 0.0162 M
- B) 0.0075 M
- C) 0.0041 M
- D) 0.0013 M

15. Suppose that MgBr_2 dissolves in pure water to give Mg^{2+} , Br^- , and MgBr^+ . The mass balance for Br^- in 0.15 M aqueous solution of MgBr_2 is:

- A) $0.45 \text{ M} = [\text{Br}^-] + [\text{MgBr}^+]$
- B) $0.45 \text{ M} = 2[\text{Br}^-] + [\text{MgBr}^+]$
- C) $0.3 \text{ M} = [\text{Br}^-] + [\text{MgBr}^+]$
- D) $0.3 \text{ M} = 2[\text{Br}^-] + [\text{MgBr}^+]$

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15
A	B	D	B	A	C	A	B	A	D	C	D	B	A	C

16. **Question 16:** Suppose that MgBr_2 dissolves in pure water to give Mg^{2+} , Br^- , and MgBr^+ . The charge balance for the solution is:

- A) $[\text{Mg}^{2+}] + [\text{H}^+] = [\text{Br}^-] + [\text{OH}^-]$
- B) $2[\text{Mg}^{2+}] + [\text{H}^+] = [\text{Br}^-] + [\text{OH}^-]$
- C) $[\text{Mg}^{2+}] + [\text{H}^+] = 2[\text{Br}^-] + [\text{OH}^-]$
- D) $2[\text{Mg}^{2+}] + [\text{H}^+] = 2[\text{Br}^-] + [\text{OH}^-]$

17. **Question 17:** Which of the following is NOT a characteristic of a buffer solution?

- A) It resists changes in pH when small amounts of acid or base are added
- B) It contains a weak acid and its conjugate base
- C) It has a pH equal to the pKa of the weak acid
- D) It can be prepared by mixing a strong acid with a strong base

18. **Question 18:** What is the pH of a solution prepared by mixing 50.0 mL of 0.10 M HCl with 50.0 mL of 0.10 M NaOH?

- A) 1.0
- B) 7.0
- C) 13.0
- D) 12.0

19. **Question 19:** Which of the following statements is true regarding the solubility product constant (K_{sp})?

- A) It is temperature-dependent
- B) It is independent of the concentration of the ions in solution
- C) It is the same for all salts of the same stoichiometry
- D) It decreases with increasing temperature

20. **Question 20:** What is the pH of a 0.10 M solution of a weak acid (HA) with a K_a of 1.0×10^{-5} ?

- A) 2.0
- B) 3.0
- C) 4.0
- D) 5.0

Q16	Q17	Q18	Q19	Q20
D	D	B	A	B